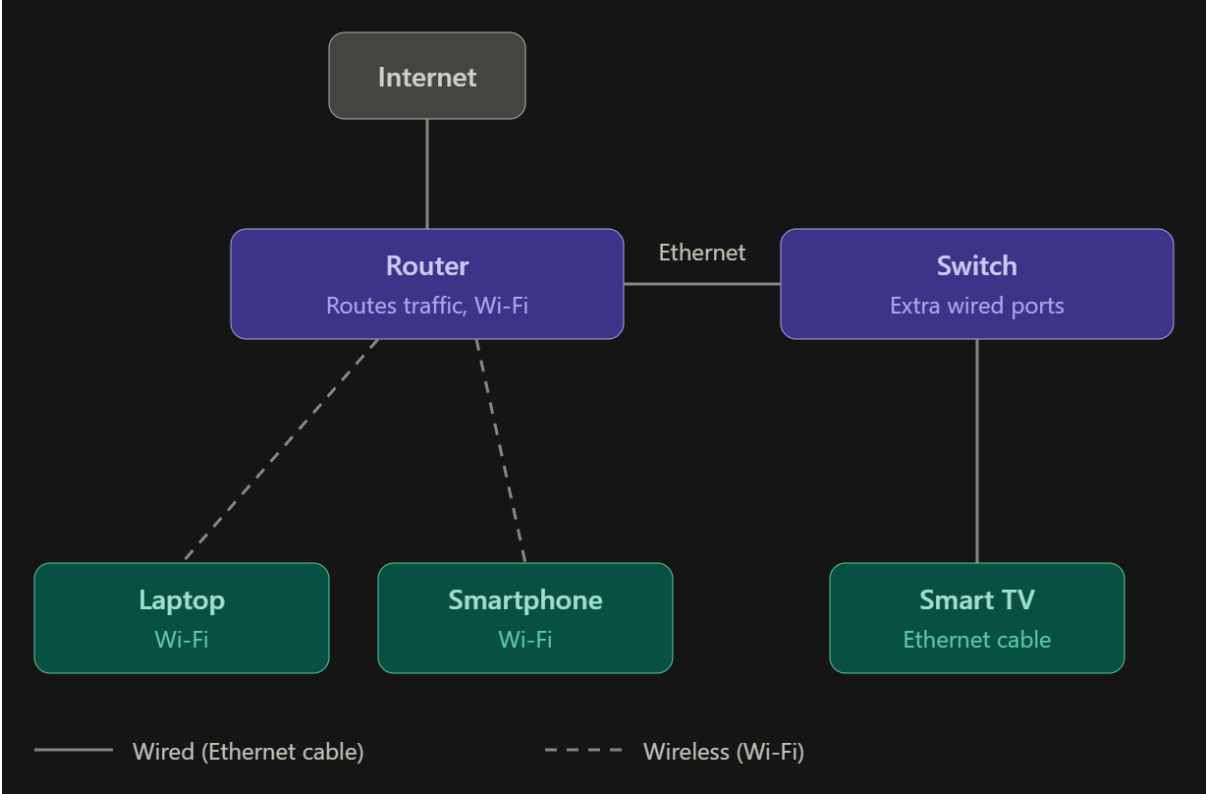


Networks Devices

1. Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.



A **router** connects the home network to the Internet and provides Wi-Fi connectivity. The router is connected to a **network switch** using an Ethernet cable. The **laptop** and **Smart TV** are connected to the switch using Ethernet cables. The **smartphone** connects wirelessly to the router using Wi-Fi.

2. Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.).

Network Device	Main Function	Real-World Use Case	Daily App/Service Example
Router	Connects different networks and provides Internet access.	Connects devices in a home to the Internet.	YouTube: A phone or laptop uses the router to stream videos.

Network Device	Main Function	Real-World Use Case	Daily App/Service Example
Switch	Connects multiple devices within the same LAN.	Connects PCs in a college computer lab or gaming café.	LAN Gaming: Multiple PCs can connect through a switch to play together.
Hub	Connects multiple devices and broadcasts data to all connected ports.	Used in simple or older small networks for connecting computers.	File Sharing: Connected computers can communicate over a basic local network.
Access Point (AP)	Provides wireless Wi-Fi access to a wired network.	Provides Wi-Fi coverage in a home, college, or office.	WhatsApp/Instagram: A smartphone connects through Wi-Fi to access these services.

3. Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).

Scenario: Ordering Food on Zomato

When I order food using the **Zomato app** on my phone, the data travels through several network devices before reaching the Internet. First, my phone connects to the **Access Point (AP)** through Wi-Fi. The access point sends the data to a **switch**, which forwards it through the local network. The data then reaches the **router**, which connects my home network to the Internet and forwards my request to Zomato's servers. Zomato processes my order and sends the response back through the Internet, router, switch, and access point to my phone.

Data path:

Phone → Access Point → Switch → Router → Internet → Zomato Server

4. Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup.

Connecting a Gaming PC to a Wi-Fi Router

If my gaming PC only has an **Ethernet port** and is located far from the Wi-Fi router, I can use the following two network devices:

Network Device	How It Works
Wi-Fi Bridge / Wireless Ethernet Bridge	The device connects wirelessly to my Wi-Fi router and provides an Ethernet port . I connect the gaming PC to the bridge using an Ethernet cable, allowing the PC to access the Internet through Wi-Fi.
Powerline Adapter	One adapter is connected to the router using Ethernet and plugged into a power socket. A second adapter is placed near the gaming PC and connected to it using Ethernet. The adapters use the home's electrical wiring to carry the network connection to the PC.

Conclusion: A **Wi-Fi bridge** is useful when a wireless connection is available, while a **powerline adapter** can be useful when I want to avoid running a long Ethernet cable through the room.